Quality, Value and Residual Momentum in US Equities

A Black-Litterman tilt on 15 size × score portfolios, 1970–2025

Project guide for the report team

Applied Investments 2026–27 (Prof. Amit Goyal), University of Lausanne

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Abstract

This document describes the whole project as implemented in the notebook `strategy.ipynb`: data, universe, signals, portfolio construction, evaluation, results and their interpretation. It gives every equation and the timing convention behind each computation, the numbers of the final run, and the file name of every figure and table, so that the report can be written without reading the code. All results are net of trading costs unless stated otherwise. All t-statistics use Newey-West standard errors with 6 lags.

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1 The project in one page

Idea. Each month we rank US common stocks *within their sector* on a composite score built from three ingredients:

- **quality**: six accounting ratios;
- **value**: book-to-market;
- **residual momentum**: past returns not explained by the Fama-French 3 factors.

The stocks are grouped into 15 value-weighted portfolios: 3 size buckets × 5 score groups. A Black-Litterman model expresses one view per size bucket: *high-score stocks beat low-score stocks.* An optimiser then tilts a long-only portfolio away from the market weights of the 15 portfolios. It must stay within ±5% of the market weight per portfolio and within a 3% tracking-error budget. The benchmark is the value-weighted portfolio of the same universe.

Key numbers Strategy: February 1970 – December 2025 (671 months). Signal: January 1966 – December 2025 (720 months), as in Table 2.

	Signal (Q5 − Q1 sort)	Strategy vs. benchmark (net)
Return (% p.a.)	5.14 ($t = 4.83$)	12.63 (benchmark 12.09)
Active return (% p.a.)	–	0.53 ($t = 2.75$), IR 0.40
FF5 alpha (% p.a.)	5.90 ($t = 5.56$)	0.32 ($t = 1.70$)
FF5 + UMD alpha (% p.a.)	4.58 ($t = 4.32$)	0.20 ($t = 1.07$)
Active return from 1996 (% p.a.)	–	0.03 ($t = 0.12$)

Main messages for the report.

1. **The signal works in the cross-section.** The score predicts returns with an FF5 alpha of 5.9% p.a. ($t = 5.6$).
   - Alphas are negative for the low-score groups and positive for the high-score groups, in every size bucket.
   - Quality is the strongest component once factors are controlled for.
   - The three components are weakly correlated.
2. **Little of it reaches the portfolio.** The ±5% active limit and the long-only constraint bind every month, and the realised tracking error is only 1.3% (budget 3%). Most of the signal's alpha sits in micro and small caps, which make up only about 8% of the benchmark.
3. **Trading costs take about half of the active return**: 0.96% p.a. gross vs. 0.53% net.
4. **The active return is mostly known factors and is concentrated before 1996.**
   - It loads on HML, and on UMD before 1996.
   - The FF5 + UMD alpha is not significant net of costs.
   - The active return is about zero from 1996, even gross of costs.
5. **A simpler portfolio does better on alpha.** Holding only the top score quintile, value-weighted, earns 1.9% p.a. active net of costs, with a significant FF5 + UMD alpha (1.8%, $t = 2.8$). It takes more risk for the same information ratio, and is not significant after 1995 either (Section 9.1).

2 Data

Source	Content used	Period
CRSP monthly stock file, CIZ format (WRDS)	Return (incl. delisting return), price, shares outstanding, dividends, SIC code, share and exchange descriptors	1963–2025
CRSP/Compustat Merged, annual (WRDS)	IB, CEQ, TXDITC, AT, REVT, COGS, DLC, DLTT, ACT, LCT	FY 1963–2026
Kenneth French Data Library	Mkt-RF, SMB, HML, RMW, CMA, UMD, RF; Fama-French 12 industry definitions	1963–2025

Layout. Every variable is stored as a matrix with stocks (PERMNO) in rows and months in columns. Index i denotes a stock and t a month-end.

Timing convention (used everywhere). A decision taken at the end of month t uses only information available at t. The resulting portfolio earns the return of month $t+1$.

2.1 CRSP cleaning

- **One row per stock and month.** When several descriptor rows exist in a month, we keep the one in force at month-end and sum the dividends.
- **Market capitalisation:** $ME_{i,t} = |P_{i,t}| \times SHROUT_{i,t}$.
- **Delisting returns.** When a stock trades until the end of month t and delists during $t+1$ without a new row, CRSP folds the delisting return into the return of t. We split it into a price return at t and a delisting return at $t+1$, so that it is not known in advance and the investor holding the stock earns it:

$$r^{\text{price}}_{i,t} = \frac{P_{i,t} + D_{i,t}}{P_{i,t-1}} - 1, \qquad r^{\text{delist}}_{i,t+1} = \frac{1 + r_{i,t}}{1 + r^{\text{price}}_{i,t}} - 1. \tag{1}$$

  This affects 2,107 stock-months.

- **Trading halts.** If a stock held at t has no return at $t+1$, it earns the move from its last price to its next available price, dividends included:

$$r_{i,t+1} = \frac{P_{i,t+k} + \sum_{s=t+1}^{t+k} D_{i,s}}{P_{i,t}} - 1, \tag{2}$$

  where $t+k$ is the next month with a price; returns inside the gap are dropped. A stock that is never priced again gets -100%. This affects 283 stock-months.

2.2 Compustat timing

- A fiscal-year report with end date FYE is usable from the end of month $FYE+6$ (so it is public), for at most 12 months. At each t the most recent usable report is taken, with all variables from the same report.
- A firm is used only from its **second** year in Compustat, to avoid backfill bias.
- Only USD-reporting firms are kept.
- Book equity: $BE = CEQ + TXDITC$.

3 Universe, benchmark, size and sectors

3.1 Investment universe $\mathcal{U}_t$

A stock is **eligible** at t if all of the following hold:

- it is a US-incorporated common stock (CIZ equivalent of share codes 10/11);
- it is listed on NYSE, AMEX or NASDAQ (exchange codes 1/2/3) and actively traded;
- $|P_{i,t}| \geq \$5$ and $ME_{i,t} > 0$.

The **universe** keeps the eligible stocks that have all the accounting data the score needs at t: IB, $BE > 0$, $AT > 0$, $REVT > 0$, COGS, debt, and, for non-financials, ACT and $LCT > 0$.

- The data contain 23,717 stocks over 756 months.
- On average 3,403 stocks are eligible and 2,602 are in the universe each month.

The return of $t+1$ is always kept, even if the stock delists during $t+1$. This avoids survivorship bias.

3.2 Benchmark

The benchmark is the value-weighted portfolio of the universe, rebalanced monthly:

$$w^B_{i,t} = \frac{ME_{i,t}}{\sum_{j \in \mathcal{U}_t} ME_{j,t}}, \qquad R^B_{t+1} = \sum_{i \in \mathcal{U}_t} w^B_{i,t}\, r_{i,t+1}. \tag{3}$$

3.3 Size buckets

- Firm size is the market cap summed over all share classes of the firm (PERMCO): $ME^{\text{firm}}_{i,t}$.
- Each month, the breakpoints b^{20}_t and b^{50}_t are the 20th and 50th percentiles of ME^{firm} among eligible NYSE firms.
- Classification:
  - micro if $ME^{\text{firm}} < b^{20}_t$;
  - small if $b^{20}_t \leq ME^{\text{firm}} < b^{50}_t$;
  - large otherwise.
- Stocks migrate between buckets over time.

3.4 Sectors

Sectors are the Fama-French 12 industries, mapped from the CRSP SIC code at t. A sector with fewer than 20 universe firms in a month is merged into *Other* for that month.

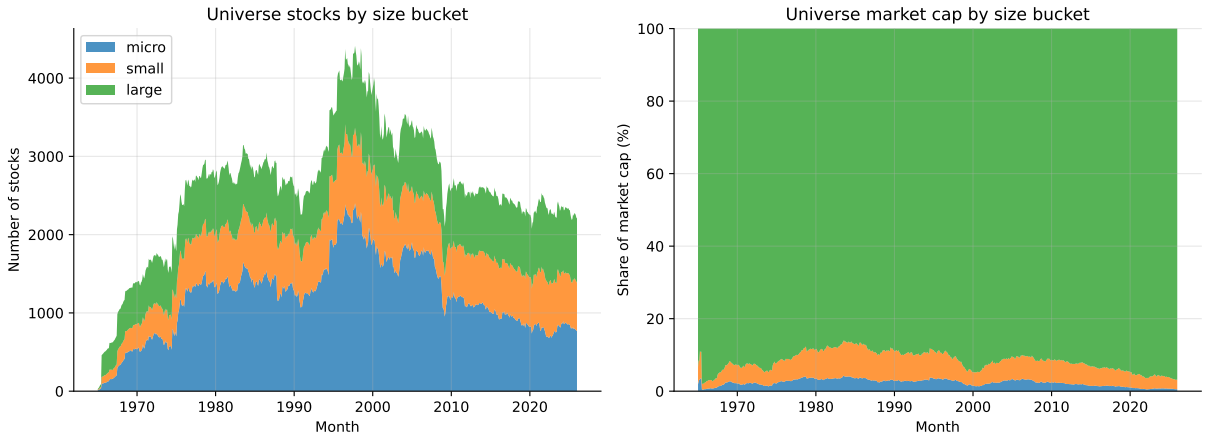


Figure 1: Universe over time by size bucket: number of stocks and share of market cap. On average 1,213 micro, 637 small and 752 large stocks; large caps make up 91.7% of market cap, small 5.9% and micro 2.3%. File: `v2_universe.pdf`.

4 Signals

4.1 Within-sector standardisation

Every raw characteristic $x_{i,t}$ is handled within its sector s and month t, in two steps:

1. it is winsorised at the 1st and 99th percentiles of the sector-month: $\tilde{x}_{i,t}$;
2. it is converted to a z-score:

$$z(x)_{i,t} = \frac{\tilde{x}_{i,t} - \bar{\tilde{x}}_{s,t}}{\sigma(\tilde{x})_{s,t}}. \tag{4}$$

Comparing stocks only with peers of the same industry removes the large level differences across sectors (Section 7).

4.2 Quality

Ratio	Definition
Return on equity	IB/BE
Return on assets	IB/AT
Gross margin	$(REVT - COGS)/REVT$
Leverage (enters with a minus)	$-(DLC + DLTT)/BE$
Current ratio (non-financials only)	ACT/LCT
Gross profitability (Novy-Marx, 2013)	$(REVT - COGS)/AT$

Quality is the mean of the K_i available ratio z-scores ($K_i = 6$, or 5 for financials, SIC 6000–6999), standardised again within sector-month:

$$Q_{i,t} = z\left(\frac{1}{K_i}\sum_{k=1}^{K_i} z(x_k)_{i,t}\right). \tag{5}$$

4.3 Value

Value is the z-score of book-to-market, with the firm-level market cap at t:

$$V_{i,t} = z\left(\frac{BE_{i,t}}{ME_{i,t}^{\text{firm}}}\right). \tag{6}$$

4.4 Residual momentum (Blitz, Huij & Martens, 2011)

At each t, for each stock with 36 months of returns, we estimate a Fama-French 3-factor regression over $\tau \in [t-35, t]$:

$$r_{i,\tau} - r_\tau^f = a_i + \beta_i^\top f_\tau + e_{i,\tau}, \qquad f_\tau = (\text{Mkt-RF}, \text{SMB}, \text{HML})_\tau. \tag{7}$$

Residual momentum is the sum of the residuals from $t-11$ to $t-1$ (the last month is skipped, to avoid short-term reversal), divided by their standard deviation:

$$RM_{i,t} = \frac{\sum_{\tau=t-11}^{t-1} \hat{e}_{i,\tau}}{\sigma(\hat{e}_{i,t-11:t-1})}. \tag{8}$$

It is designed to be less exposed to the UMD factor than plain momentum. In this sample, however, it still loses more than half of its alpha when UMD is added (Section 8). It is winsorised and z-scored within sector-month like the other signals. A stock without 36 months of history gets a neutral value of 0. The residual momentum is available for 93% of universe stock-months.

4.5 Composite score

The three components have equal weights and nothing is fitted:

$$S_{i,t} = z(Q_{i,t} + V_{i,t} + z(RM)_{i,t}). \tag{9}$$

5 Portfolio construction

5.1 Signal validation sort

Each month the universe is split into quintiles of $S_{i,t}$, with $G_{i,t} = \lceil 5 \cdot \text{rank}_{\%}(S_{i,t}) \rceil$. The value-weighted long-short return is

$$R_{t+1}^{Q5-Q1} = \sum_{i \in G=5} \frac{ME_{i,t}}{\sum_{j \in G=5} ME_{j,t}} r_{i,t+1} - \sum_{i \in G=1} \frac{ME_{i,t}}{\sum_{j \in G=1} ME_{j,t}} r_{i,t+1}. \tag{10}$$

This sort is a diagnostic of the signal only. It is not the strategy. The sorts in Table 2 use the months where all four sorts exist, January 1966 – December 2025 (residual momentum needs 36 months of returns). They therefore start earlier than the strategy (February 1970).

5.2 The 15 portfolios

Within each size bucket b, stocks are sorted into score quintiles $q = 1, \ldots, 5$. Each cell $c = (b, q)$ is a value-weighted portfolio:

$$R_{c,t+1} = \frac{\sum_{i \in c_t} ME_{i,t}\, r_{i,t+1}}{\sum_{i \in c_t} ME_{i,t}}, \qquad w_{c,t}^{mkt} = \frac{\sum_{i \in c_t} ME_{i,t}}{\sum_{i \in \mathcal{U}_t} ME_{i,t}}. \tag{11}$$

Held at w_t^{mkt}, the 15 portfolios reproduce the benchmark exactly. The notebook checks this every month. Working with 15 portfolios instead of about 2,600 stocks makes the covariance matrix estimable ($N = 15 \ll T = 60$).

5.3 Covariance

Σ_t is the sample covariance of the 15 portfolio returns over 60 months. The window ends with the return of month $t-1$: the return of month t is left out, a conservative one-month gap.

5.4 Black-Litterman

Prior. The equilibrium (implied) returns are

$$\Pi_t = \delta\, \Sigma_t\, w_t^{mkt}, \qquad \delta = 2.5. \tag{12}$$

Views. There is one view per size bucket, $k \in \{\text{micro}, \text{small}, \text{large}\}$.

- Row k of P (3×15) puts grades $(-1, -\frac{1}{2}, 0, \frac{1}{2}, 1)$ on Q1–Q5 of bucket k and zero elsewhere. The view portfolio is therefore $Q5 - Q1 + \frac{1}{2}(Q4 - Q2)$, so the middle groups carry information too.
- The view's expected return is $q_k = 4\%$ p.a. ($4\%/12$ per month).
- The view uncertainty is $\Omega = \text{diag}(P\, \tau\Sigma_t\, P^\top)$, with $\tau = 1/60$.

Posterior mean.

$$\mu_t = \Pi_t + \tau\Sigma_t P^\top \left(P\, \tau\Sigma_t\, P^\top + \Omega\right)^{-1} (q - P\, \Pi_t)\,. \tag{13}$$

5.5 Optimisation

The weights maximise the Black-Litterman alpha of the active position, in the spirit of Grinold & Kahn (2000):

$$\begin{aligned} \max_{w} \quad & (\mu_t - \Pi_t)^\top (w - w_t^{mkt}) \\ \text{s.t.} \quad & \mathbf{1}^\top w = 1, \qquad w \geq 0 \;\; \text{(long-only)}, \\ & |w_c - w_{c,t}^{mkt}| \leq 5\% \quad \forall c, \\ & \sqrt{12\,(w - w_t^{mkt})^\top \Sigma_t\, (w - w_t^{mkt})} \leq 3\%. \end{aligned} \tag{14}$$

The problem is solved each month with cvxpy (Clarabel solver). As a sanity check, a mean-variance optimiser fed with Π_t and no views must return w_t^{mkt}. The strategy starts in January 1970, the first month with 60 months of portfolio returns, and earns its first return in February 1970.

5.6 Stock weights, turnover and trading costs

- **Stock weights.** Each stock gets its portfolio's weight times its value weight inside the portfolio:

$$w_{i,t} = W_{c(i),t} \cdot \frac{ME_{i,t}}{\sum_{j \in c(i)} ME_{j,t}}.$$

- **Drifted weights.** Between rebalancings, the weights of $t-1$ drift with the returns of t:

$$\tilde{w}_{i,t} = \frac{w_{i,t-1}(1 + r_{i,t})}{1 + \sum_j w_{j,t-1}\, r_{j,t}}.$$

- **Costs.** One-way cost c_i of 50 / 25 / 10 bps for micro / small / large stocks. The cost of rebalancing at t is paid out of the return of $t+1$:

$$TC_t = \sum_i c_i \, |w_{i,t} - \tilde{w}_{i,t}|, \qquad R^{S,\text{net}}_{t+1} = \sum_c W_{c,t} R_{c,t+1} - TC_t. \tag{15}$$

- **Turnover**, one-way: $\frac{1}{2}\sum_i |w_{i,t} - \tilde{w}_{i,t}|$.
- The benchmark pays the same costs on its own trades.

6 Evaluation

6.1 Statistics

Let R_t be the monthly strategy return, R^B_t the benchmark return, r^f_t the risk-free rate and $A_t = R_t - R^B_t$ the active return.

Statistic	Definition
Return, volatility	$12\,\overline{R}$, $\sqrt{12}\,\sigma(R)$
Sharpe ratio	$\sqrt{12}\,\overline{R - r^f} \,/\, \sigma(R - r^f)$
Sortino ratio	$\sqrt{12}\,\overline{R - r^f} \,/\, \sqrt{\overline{\min(R - r^f, 0)^2}}$
CAGR	$(\prod_t (1 + R_t))^{12/T} - 1$
Max drawdown	$\min_t \left(V_t / \max_{s \le t} V_s - 1\right)$, $V_t = \prod_{s \le t}(1 + R_s)$
Calmar ratio	CAGR / |max drawdown|
Longest drawdown	longest run of months with V_t below its previous peak
VaR / CVaR 5%	5th percentile of R_t / mean of R_t below it (monthly)
Beta	$\mathrm{cov}(R, R^B)/\mathrm{var}(R^B)$
Active return, tracking error	$12\,\overline{A}$, $\sqrt{12}\,\sigma(A)$
Information ratio	$\sqrt{12}\,\overline{A}/\sigma(A)$
Hit rate	share of months with $R_t > R^B_t$

All statistics are reported **net** and **gross** of trading costs. They are given for the whole period and for two halves: up to December 1995, and from January 1996.

6.2 Factor regressions

The active return is regressed on four factor models: CAPM (Mkt-RF), FF3 (Fama & French, 1993), FF5 (Fama & French, 2015) and FF5 + UMD (Carhart, 1997):

$$A_t = \alpha + \beta^\top F_t + \varepsilon_t. \tag{16}$$

The two other regressions use the same form:

- the long-short signal returns R^{Q5-Q1}_t are regressed directly on the factors;
- the 15 portfolios are regressed with their excess return $R_{c,t} - r^f_t$ on the left-hand side.

Alphas are annualised ($\times 12$). FF3 uses the 3-factor file of the French library, whose SMB differs from the FF5 one. Standard errors are Newey-West (1987) with 6 lags.

7 Descriptive statistics

7.1 Ratios by sector

Table 1: Median of each characteristic by sector, all universe stock-months, before winsorisation. File: `v2_ratios_by_sector.tex`.

Sector	Obs. (%)	ROE	ROA	GM	D/E	CR	GP	B/M	RM
BusEq	13.3	0.09	0.05	0.46	0.15	2.65	0.41	0.37	−0.14
Chems	2.9	0.12	0.06	0.35	0.45	2.21	0.37	0.52	−0.12
Durbl	3.1	0.12	0.06	0.27	0.37	2.36	0.38	0.62	−0.09
Enrgy	3.8	0.09	0.05	0.36	0.51	1.42	0.20	0.59	−0.03
Hlth	6.9	0.08	0.04	0.48	0.21	3.04	0.36	0.28	−0.17
Manuf	14.3	0.11	0.06	0.28	0.42	2.35	0.33	0.67	−0.08
Money	18.5	0.10	0.01	0.49	0.87	–	0.04	0.72	−0.01
NoDur	7.4	0.11	0.06	0.33	0.43	2.28	0.43	0.65	−0.05
Other	12.1	0.10	0.05	0.28	0.46	1.69	0.26	0.51	−0.11
Shops	10.8	0.12	0.05	0.29	0.45	1.90	0.53	0.55	−0.14
Telcm	2.0	0.07	0.03	0.49	0.79	1.46	0.21	0.49	−0.11
Utils	4.9	0.10	0.04	0.30	1.00	0.89	0.11	0.89	0.05
All	100.0	0.10	0.04	0.35	0.47	2.10	0.29	0.58	−0.08

GM: gross margin; D/E: debt to equity; CR: current ratio; GP: gross profitability; B/M: book-to-market; RM: residual momentum.

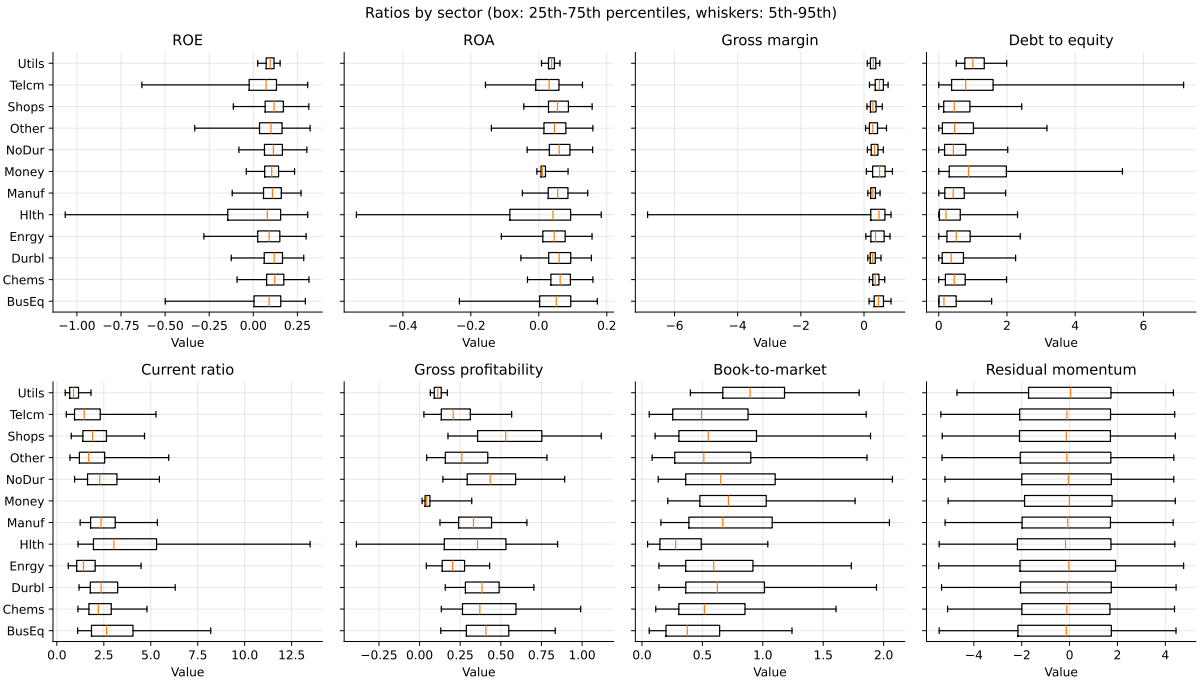


Figure 2: Distribution of each characteristic by sector (box: 25th–75th percentiles, whiskers: 5th–95th). File: `v2_ratios_by_sector.pdf`.

Reading. Levels differ strongly across industries. For example:

- debt to equity: 0.15 in business equipment vs. 1.00 in utilities;
- gross profitability: 0.04 in finance vs. 0.53 in retail;
- health has a very wide spread of ROE and margins.

A raw ranking would mostly sort industries, which is why every signal is standardised within sector.

7.2 Whole universe and correlations

	Coverage (%)	P5	P25	Median	P75	P95
ROE	100.0	−0.26	0.05	0.10	0.15	0.28
ROA	100.0	−0.11	0.01	0.04	0.08	0.15
Gross margin	100.0	0.08	0.23	0.35	0.51	0.80
Debt to equity	100.0	0.00	0.14	0.47	1.00	3.13
Current ratio	81.5	0.76	1.42	2.10	3.09	6.60
Gross profitability	100.0	0.03	0.12	0.29	0.47	0.82
Book-to-market	100.0	0.10	0.32	0.58	0.94	1.79
Residual momentum	92.8	−5.25	−2.02	−0.08	1.74	4.40

The current ratio is not defined for financials, hence its coverage of 81.5%. The average monthly cross-sectional correlations of the three scores are low:

- quality / value: −0.25;
- quality / residual momentum: −0.09;
- value / residual momentum: −0.06.

Quality firms tend to be expensive, so quality and value partly hedge each other, and combining the three components diversifies the signal.

8 Results

8.1 Does the score predict returns?

Table 2: Value-weighted Q5 − Q1 sorts on each component and on the score, January 1966 – December 2025, % p.a. (t). File: `v2_signals.tex`.

Signal	Q5 − Q1	FF5 alpha	FF5 + UMD alpha
Quality	0.64 (0.42)	3.98 (3.89)	3.39 (3.39)
Value	1.93 (0.98)	−1.09 (−0.86)	1.33 (1.20)
Residual momentum	4.10 (3.42)	4.43 (3.41)	1.88 (1.56)
Score	**5.14 (4.83)**	**5.90 (5.56)**	**4.58 (4.32)**

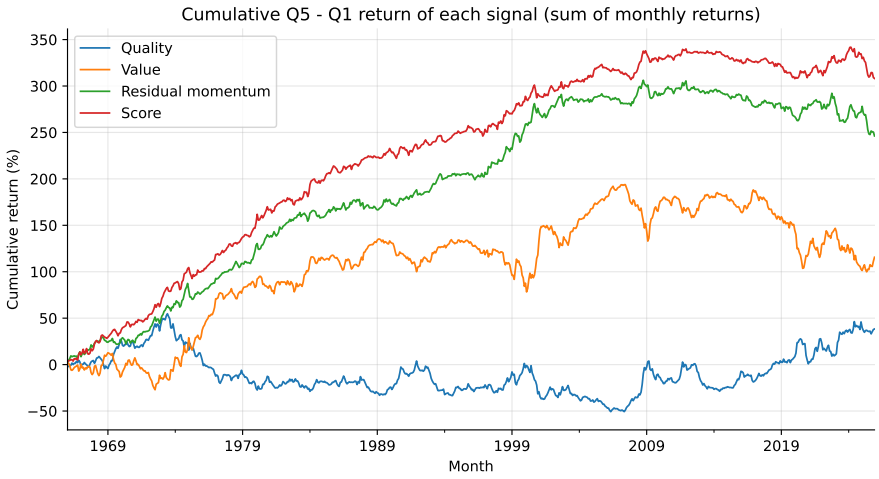


Figure 3: Cumulative Q5 − Q1 return of each signal (sum of monthly returns). File: `v2_signal_spreads.pdf`.

Reading.

- **Quality** has almost no raw spread, because high-quality stocks are expensive and so load negatively on HML. Once the factors are controlled for, it has the largest alpha, and that alpha survives UMD.
- **Value** alone adds nothing beyond FF5: it *is* HML.
- **Residual momentum** loses more than half of its alpha when UMD is added (4.43% to 1.88%): it is closer to plain momentum than its design suggests.
- **The composite score** has a larger and more significant alpha than any component, a direct benefit of the low correlations.

The spreads flatten from the mid-2000s (figure).

8.2 The 15 size × score portfolios

Table 3: Characteristics of the 15 portfolios (monthly averages). Return of $t + 1$; FF5 alpha of the excess return, % p.a. File: `v2_portfolios.tex`.

Portfolio	Stocks	Median ME ($m)	Bench. weight (%)	Return	Vol.	FF5 alpha	t
micro-Q1	242	132	0.50	7.06	23.03	−6.70	−8.55
micro-Q2	243	134	0.50	11.81	21.23	−2.56	−3.58
micro-Q3	243	125	0.48	15.18	20.85	0.47	0.65
micro-Q4	243	112	0.46	17.41	20.73	2.91	4.15
micro-Q5	243	92	0.40	19.76	20.92	4.49	4.84
small-Q1	127	699	1.19	9.51	21.70	−4.26	−5.86
small-Q2	127	711	1.21	12.67	20.30	−1.32	−2.21
small-Q3	127	697	1.20	14.02	19.80	−0.18	−0.34
small-Q4	127	684	1.18	16.36	19.69	2.24	3.74
small-Q5	128	666	1.17	17.56	19.93	2.89	4.37
large-Q1	150	3,900	17.09	9.36	16.93	−2.52	−4.12
large-Q2	150	4,030	18.03	10.88	16.11	−0.76	−1.21
large-Q3	150	3,994	18.68	10.28	15.14	−0.85	−1.54
large-Q4	150	3,909	19.06	11.91	15.38	0.84	1.44
large-Q5	151	3,547	18.85	14.13	15.44	3.42	5.25

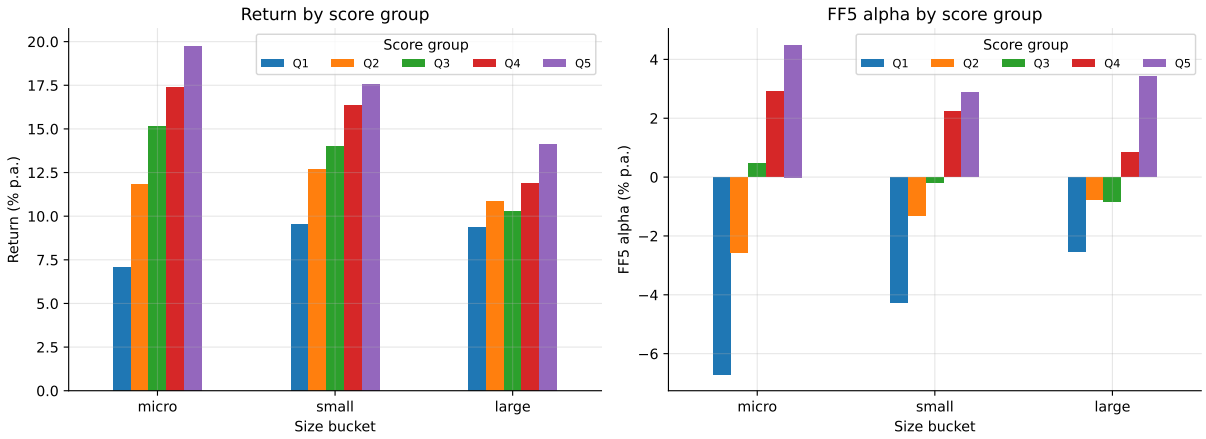


Figure 4: Average return and FF5 alpha by score group within each size bucket. File: `v2_portfolios.pdf`.

Reading. In every size bucket the alpha rises monotonically from Q1 to Q5.

- It is negative for the low-score groups and positive for the high-score groups, so the signal works on both sides: avoiding the bad stocks pays as much as holding the good ones.
- The Q5 − Q1 alpha gap is about 11 points in micro caps, 7 in small and 6 in large: most of the signal is in the smallest stocks.
- Micro and small caps together weigh only about 8% of the benchmark, which limits how much of this alpha a benchmark-relative portfolio can reach.

8.3 Implementation

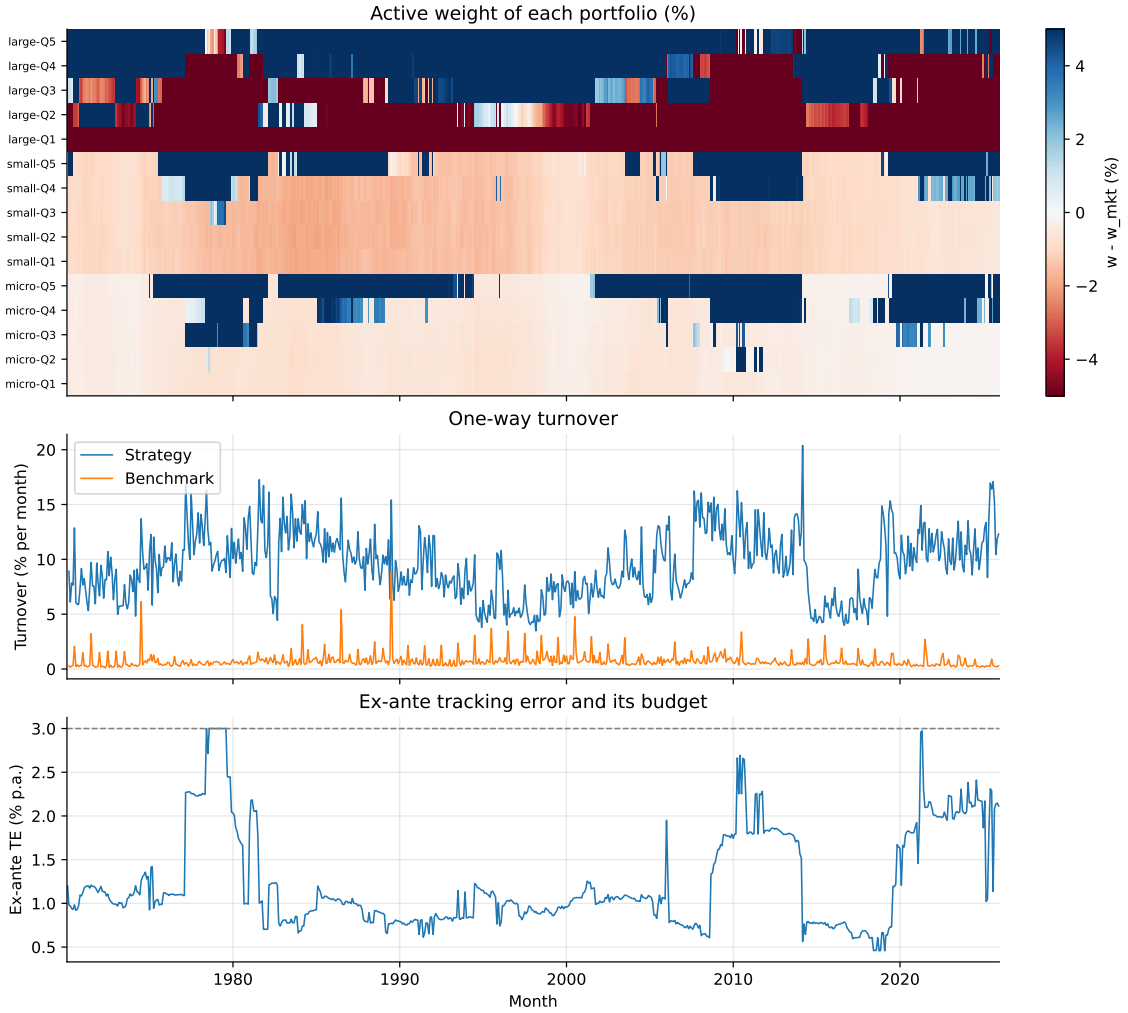


Figure 5: Active weight of each portfolio over time, one-way turnover, and ex-ante tracking error vs. its 3% budget. File: `v2_implementation.pdf`.

Average active bets (% of the portfolio, $w - w^{mkt}$):

	Q1	Q2	Q3	Q4	Q5
micro	−0.5	−0.4	0.1	1.4	3.2
small	−1.2	−1.3	−1.2	0.2	2.0
large	−5.0	−3.4	−0.4	1.9	4.5

Reading.

- **Every month** at least one portfolio is at the ±5% active limit, and at least one is at zero weight (long-only).
- The low-score micro and small portfolios cannot be underweighted below zero.
- The ex-ante tracking error averages 1.24% and reaches the 3% budget in only 2% of months. The **binding constraints are the active limit and long-only**, not the risk budget.
- The high-score micro portfolio is held at about 3.6% of the portfolio on average, roughly 9 times its 0.4% benchmark weight. The portfolio still stays about 89% in large caps.
- Turnover is 9.4% per month (benchmark 0.7%), and trading costs are 0.45% p.a. (benchmark 0.03%).

8.4 Performance

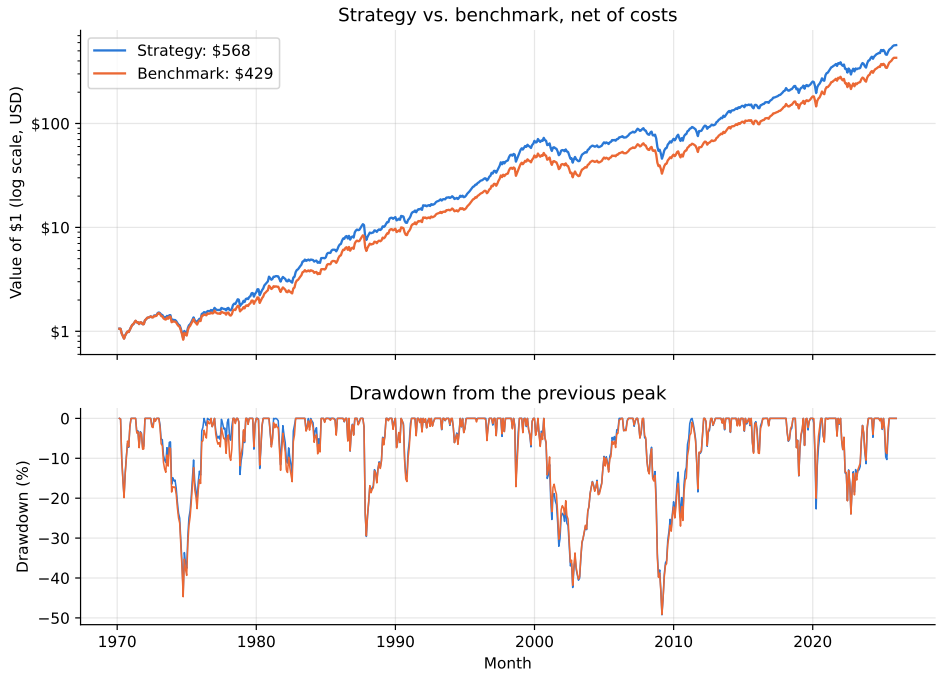


Figure 6: Value of $1 invested (log scale), net of costs, and drawdowns. File: `v2_strategy_vs_benchmark.pdf`.

Table 4: Performance, net and gross of costs. File: `v2_performance.tex`.

	Whole		Up to 1995		From 1996	
	Net	Gross	Net	Gross	Net	Gross
Return (% p.a.)	12.63	13.08	14.02	14.48	11.42	11.87
Volatility (%)	15.62	15.62	15.60	15.60	15.66	15.65
Sharpe	0.53	0.56	0.46	0.49	0.58	0.61
Max drawdown (%)	−49.24	−48.73	−43.61	−43.42	−49.24	−48.73
Active (% p.a.)	0.53	0.96	1.11	1.54	0.03	0.46
t (active)	2.75	4.86	4.17	5.48	0.12	1.78
Tracking error (%)	1.33	1.34	1.21	1.22	1.42	1.42
Information ratio	0.40	0.72	0.92	1.26	0.02	0.32
Benchmark return (% p.a.)	12.09	12.12	12.91	12.94	11.39	11.41

Table 5: Other risk statistics: strategy net, gross, and benchmark (net). File: `v2_risk.tex`.

	Whole			Up to 1995			From 1996		
	Net	Gross	Bench.	Net	Gross	Bench.	Net	Gross	Bench.
CAGR (%)	12.01	12.51	11.45	13.58	14.09	12.34	10.67	11.17	10.68
Sortino	0.78	0.83	0.74	0.69	0.74	0.58	0.86	0.91	0.87
Calmar	0.24	0.26	0.23	0.31	0.32	0.28	0.22	0.23	0.22
Longest DD (months)	64	64	66	38	38	44	64	64	66
Skewness	−0.51	−0.51	−0.49	−0.45	−0.45	−0.37	−0.56	−0.56	−0.60
Excess kurtosis	1.93	1.93	1.72	3.04	3.03	2.63	1.03	1.03	0.93
VaR 5% (%/month)	−7.00	−6.96	−6.92	−5.71	−5.65	−5.71	−7.78	−7.70	−7.41
CVaR 5% (%/month)	−9.92	−9.87	−9.74	−9.61	−9.57	−9.42	−9.93	−9.88	−9.79
Beta	1.01	1.01	–	1.00	1.00	–	1.01	1.01	–
Hit rate (%)	56.5	60.7	–	65.3	68.8	–	48.9	53.6	–

Reading.

- The strategy is a market-like portfolio (beta 1.01, tracking error 1.3%) with a small positive active return over the whole sample.
- Its risk profile is almost identical to the benchmark's: drawdowns, skewness and tail risk barely differ.
- The active return is significant up to 1995 (1.11% net, IR 0.92) and vanishes afterwards (0.03% net, IR 0.02). From 1996 the strategy beats the benchmark in fewer than half of the months, net of costs.

8.5 Alpha and factor exposures

Table 6: Alpha of the active return (strategy − benchmark), % p.a. (t). File: `v2_alpha.tex`.

	Whole		Up to 1995		From 1996	
	Net	Gross	Net	Gross	Net	Gross
CAPM	0.49 (2.41)	0.91 (4.47)	1.13 (4.33)	1.55 (5.70)	−0.09 (−0.31)	0.34 (1.21)
FF3	0.31 (1.59)	0.74 (3.76)	0.97 (3.73)	1.40 (5.15)	−0.17 (−0.62)	0.26 (0.95)
FF5	0.32 (1.70)	0.75 (3.96)	0.82 (2.82)	1.25 (4.17)	−0.13 (−0.48)	0.29 (1.09)
FF5 + UMD	0.20 (1.07)	0.63 (3.35)	0.41 (1.50)	0.84 (2.96)	−0.14 (−0.52)	0.29 (1.07)

Table 7: FF5 + UMD loadings of the net active return, coefficient (t). File: `v2_loadings.tex`.

	Whole	Up to 1995	From 1996
Alpha (% p.a.)	0.20 (1.07)	0.41 (1.50)	−0.14 (−0.52)
Mkt-RF	0.010 (2.28)	0.002 (0.24)	0.011 (1.93)
SMB	0.017 (1.40)	0.020 (1.21)	0.022 (1.44)
HML	0.047 (4.68)	0.024 (1.78)	0.051 (4.83)
RMW	−0.001 (−0.05)	−0.001 (−0.07)	0.001 (0.11)
CMA	−0.011 (−0.88)	0.014 (0.79)	−0.021 (−1.60)
UMD	0.015 (2.27)	0.045 (4.76)	0.001 (0.27)
R^2	0.13	0.23	0.19

Reading.

- **Costs.** Gross of costs, the active return has a significant alpha under every model over the whole sample: FF5 + UMD 0.63%, $t = 3.4$. Net of costs, only the CAPM alpha stays significant. Trading costs therefore remove the significance over the whole sample.
- **Factor exposures.** Part of the active return is explained by known factors: HML (value) over the whole sample, and UMD (momentum) before 1996.
- **After 1995.** No alpha is significant, net or gross: the signal itself fades, independently of costs.

8.6 Active return over time

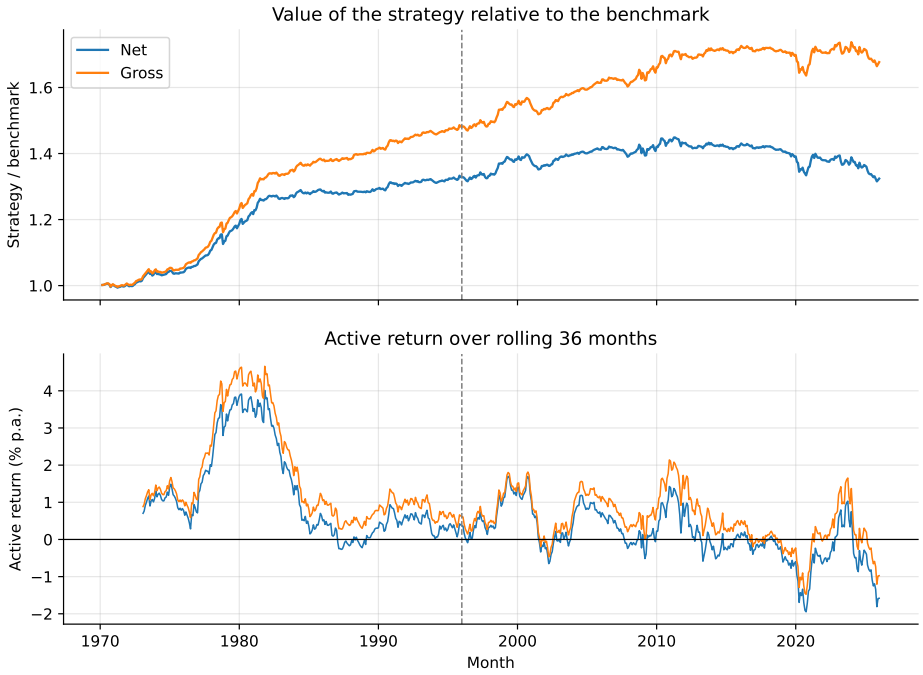


Figure 7: Value of the strategy relative to the benchmark (net and gross) and active return over rolling 36 months. The dashed line marks the end of 1995. File: `v2_active_return.pdf`.

Table 8: Active return by decade and market state, % p.a. (t). File: `v2_regimes.tex`.

	Months	Net	Gross
1970s	119	1.77 (3.56)	2.16 (4.03)
1980s	120	0.88 (2.24)	1.40 (3.45)
1990s	120	0.65 (2.14)	0.90 (3.04)
2000s	120	0.29 (0.81)	0.72 (1.88)
2010s	120	−0.18 (−0.66)	0.26 (0.93)
2020s	72	−0.73 (−0.83)	−0.11 (−0.12)
Benchmark up	419	0.49 (1.74)	0.91 (3.15)
Benchmark down	252	0.61 (1.44)	1.03 (2.52)

Reading.

- The active return declines decade after decade. Most of the gain was made in the 1970s and early 1980s, and the relative value has been flat to slightly falling since about 2000.
- The strategy does about as well in up and down markets, so it is not a hidden market-timing bet.
- This pattern matches the post-publication decay of anomaly returns documented by McLean & Pontiff (2016): value, momentum and profitability were all documented between the early 1990s and early 2010s.

9 Robustness

The backtest is re-run with one change at a time. The variants are reported for information and do not change the main specification. In Tables 9 and 10, *Active* is the mean of strategy − benchmark, as in Table 4. It is not a factor alpha: the alphas are in Table 6 (for example, the net CAPM alpha is 0.49% while the net active return is 0.53%).

Table 9: Robustness of the net active return, % p.a. (t). File: `v2_robustness.tex`.

Variant	Active	Active from 1996	TE (%)	FF5 + UMD alpha	Short (%)
Base	**0.53 (2.75)**	**0.03 (0.12)**	**1.33**	**0.20 (1.07)**	0
Active limit 2.5%	0.24 (2.17)	−0.07 (−0.49)	0.76	0.08 (0.73)	0
Active limit 10%	0.91 (2.90)	0.20 (0.45)	2.08	0.44 (1.55)	0
TE budget 2%	0.51 (2.77)	0.06 (0.22)	1.26	0.22 (1.26)	0
Ledoit-Wolf covariance	0.58 (3.19)	0.13 (0.49)	1.25	0.28 (1.64)	0
Rolling view	0.35 (1.43)	−0.30 (−0.91)	1.65	−0.02 (−0.10)	0
Short selling allowed	0.64 (2.20)	−0.33 (−0.89)	1.98	0.37 (1.44)	22.2

- **Active limit.** Scaling the limit scales the active return almost proportionally, together with the tracking error. The limit is the binding constraint, as Section 8.3 showed.
- **TE budget.** Cutting the budget to 2% changes nothing: the budget does not bind.
- **Ledoit-Wolf** shrinkage of Σ (Ledoit & Wolf, 2004) helps slightly. It is the only variant with a borderline significant FF5 + UMD alpha ($t = 1.64$).
- **Rolling view.** Replacing the fixed 4% view by the 60-month average of the realised view-portfolio return weakens the strategy, because the view then follows the decay of the signal.
- **Short selling.** Dropping the long-only constraint (same ±5% limit and TE budget) leads to 22% of short positions on average, in low-score micro and small caps. The active return barely moves (0.64% vs. 0.53%) and is worse after 1995. Borrowing costs are not modelled, so even this small gain is optimistic.
- No variant gives a significant FF5 + UMD alpha at 5%, or a significant active return after 1995.

9.1 Simple long-only alternatives

Without Black-Litterman, a simple rule holds only the best score groups. For a set G of the 15 portfolios (Q5, or Q4 and Q5; all sizes, or large caps only), the stocks are value-weighted:

$$w_{i,t} = \frac{ME_{i,t}}{\sum_{j \in G_t} ME_{j,t}} \quad (i \in G_t), \qquad w_{i,t} = 0 \text{ otherwise.} \tag{17}$$

These portfolios pay the same trading costs as the strategy and are evaluated in the same way.

Table 10: Simple long-only alternatives vs. the Black-Litterman strategy, % p.a. (t). Active returns are against the benchmark (mean of portfolio − benchmark, not an alpha). File: `v2_simple.tex`.

Portfolio	Ret. net	Active net	Active gross	From 1996 net	TE	IR	FF5+UMD α net	Turn.	Costs
Black-Litterman	12.63	0.53 (2.75)	0.96 (4.86)	0.03 (0.12)	1.33	0.40	0.20 (1.07)	9.4	0.45
Q5 (all sizes)	14.01	1.92 (3.08)	2.82 (4.52)	0.79 (0.93)	4.77	0.40	1.77 (2.77)	33.0	0.93
Q4 + Q5 (all sizes)	13.20	1.10 (2.74)	1.72 (4.25)	0.21 (0.39)	3.06	0.36	0.97 (2.27)	22.6	0.64
Large Q5	13.85	1.76 (2.59)	2.54 (3.74)	0.76 (0.83)	5.21	0.34	1.75 (2.59)	33.6	0.81
Large Q4 + Q5	12.98	0.89 (2.01)	1.42 (3.20)	0.14 (0.24)	3.47	0.26	0.87 (1.94)	22.9	0.55

TE: tracking error (%); IR: information ratio; Turn.: one-way turnover (% per month); Costs: % p.a.

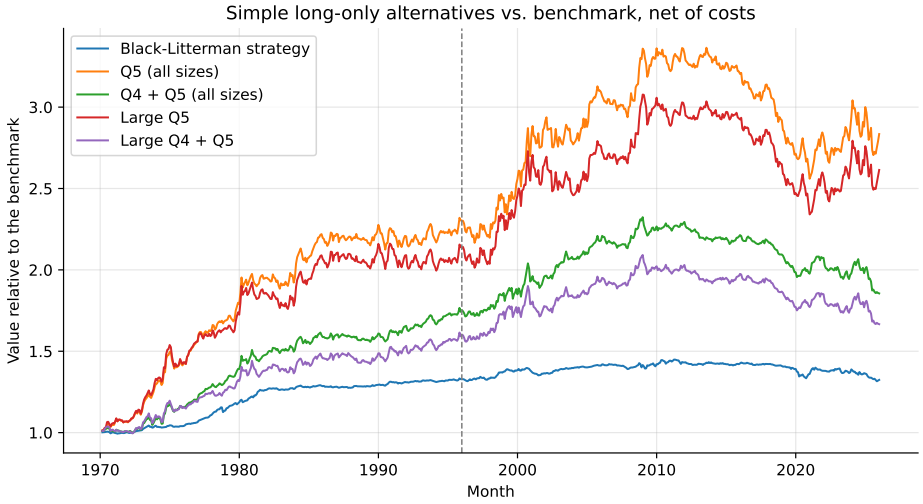


Figure 8: Value of each portfolio relative to the benchmark, net of costs. The dashed line marks the end of 1995. File: `v2_simple_alternatives.pdf`.

Reading.

- **More alpha.** Holding only Q5 earns about 3–4 times the active return of the Black-Litterman strategy (1.92% vs. 0.53% net). Its FF5 + UMD alpha is significant net of costs (1.77%, $t = 2.77$), which the Black-Litterman strategy never achieves.
- **Not a microcap effect.** The large-cap version, about 150 stocks and 19% of the market, does almost as well (1.75%, $t = 2.59$).
- **Same efficiency.** The information ratio is the same (0.40). The simple rule takes 3–4 times more tracking error (4.8% vs. 1.3%) and pays twice the costs (turnover of 33% per month).
- **Black-Litterman versus the simple rule.** The Black-Litterman implementation is a more diversified, lower-risk way to hold the same signal, not a more efficient one.
- **Adding Q4** dilutes the signal: the active return and the alpha roughly halve.
- **After 1995, none is significant.** Q5 kept outperforming until about 2010, then lost a large part of its lead (figure). This confirms the decay of the signal.

10 Discussion and limitations

10.1 Bias controls

Risk	Treatment
Look-ahead	Accounting data lagged 6 months; every estimate at t uses data up to t only; weights set at t earn the return of $t+1$; covariance window ends at $t-1$.
Survivorship	Delisted firms kept; delisting returns moved to $t+1$; held stocks never priced again get -100%.
Backfill	Firm used from its second year in Compustat.
Microcaps	\$5 price filter at t; NYSE size breakpoints; value weights.
Covariance	15 portfolios instead of thousands of stocks ($N = 15 \ll T = 60$), so Σ is invertible.
Data snooping	Equal-weighted score with no fitted weights; results reported by sub-period, by decade and with robustness variants.

10.2 Limitations to state in the report

- **Specification chosen on the full sample.** Several design choices were made after looking at full-sample results:
  - the score components;
  - the graded views and the 4% view;
  - the alpha-maximising optimiser.

  The in-sample period therefore overlaps with the evaluation period. The sub-period results are the most informative test, and they show no outperformance after 1995.
- **Trading costs** are a fixed cost per trade by size bucket. Market impact on illiquid micro-caps, and borrowing costs in the short-selling variant, are not modelled.
- **Compustat link.** The link filters used in the CRSP/Compustat export (link type, primary link) are not in the data file and cannot be checked.
- The **\$1 price filter** (instead of \$5) has not been tested.

10.3 Overall interpretation

The project separates two questions that are often mixed up:

1. **Is there information in the signal?** Yes. The quality–value–residual-momentum score has a strong and significant alpha in the cross-section, monotonic across score groups and robust to the FF5 + UMD factors.
2. **Can a realistic benchmark-relative portfolio harvest it?** Only marginally. The alpha sits mostly in small stocks that are a tiny part of the benchmark. The long-only constraint and the active limit block the short side. Costs absorb about half of what is left. And the signal has weakened sharply since the mid-1990s.

In the language of Grinold & Kahn, the *information coefficient* of the signal is good, but the *transfer coefficient* of the implementation is low. A simple rule that just holds the top quintile

transfers more of the signal (a significant alpha net of costs), at the price of more tracking error, with the same information ratio. On top of that, the signal's alpha has decayed over time.

11 Suggested report outline

1. **Introduction**: objective, the idea of combining quality, value and residual momentum, main findings (Section 1).
2. **Data and universe**: Sections 2–3, with the universe figure.
3. **Signals**: definitions and equations (Section 4), ratios by sector (table and figure), correlations.
4. **Signal evidence**: Q5 − Q1 table and figure, 15-portfolio alphas.
5. **Portfolio construction**: 15 portfolios, Black-Litterman, optimisation, costs (Section 5).
6. **Results**: performance and risk tables, \$1 chart, alpha table and loadings, active return over time and by decade.
7. **Implementation analysis**: binding constraints, turnover and costs (implementation figure).
8. **Robustness and simple alternatives**: the robustness table, and the comparison with holding only Q5 or Q4 + Q5.
9. **Discussion and limitations**: Section 10.
10. **Conclusion.**

12 Where to find everything

Topic	Notebook section	Output files (`results/`)
CRSP cleaning, delistings, halts	1–3	–
Compustat, factors	4–5	–
Universe, benchmark, size	6–8b	`figures/v2_universe.pdf`
Sectors, residual momentum	9–9b	–
Scores, descriptive statistics	10–10b	`tables/v2_ratios_by_sector.tex`, `figures/v2_ratios_by_sector.pdf`
Signal sort and decomposition	11–11b	`tables/v2_signals.tex`, `figures/v2_signal_spreads.pdf`
15 portfolios	12–12b	`tables/v2_portfolios.tex`, `figures/v2_portfolios.pdf`
Black-Litterman, optimisation	13	–
Costs, turnover, limits	14–14b	`figures/v2_implementation.pdf`
Performance and risk	15–15b	`tables/v2_performance.tex`, `tables/v2_risk.tex`
Alpha and loadings	16–16b	`tables/v2_alpha.tex`, `tables/v2_loadings.tex`
Charts over time	17–17c	`figures/v2_strategy_vs_benchmark.pdf`, `figures/v2_active_return.pdf`, `tables/v2_regimes.tex`
Robustness	18	`tables/v2_robustness.tex`
Simple alternatives	18b	`tables/v2_simple.tex`, `figures/v2_simple_alternatives.pdf`

Every figure exists as `.pdf`, for the report, and as `.png`. The `.tex` tables can be included in Overleaf with `\input{...}`; they need the `booktabs` package.

Parameter	Value	Notebook
Price filter	\$5	`MIN_PRICE`
Size breakpoints (NYSE)	20th / 50th percentile	`BREAKPOINTS`
Min. firms per sector	20	`MIN_SECTOR`
Winsorisation	1% / 99%	`WINSOR`
Residual momentum	36-month betas, $t-11$ to $t-1$	`RM_*`
Covariance window	60 months	`WINDOW`
Risk aversion / prior uncertainty	$\delta = 2.5$, $\tau = 1/60$	`DELTA`, `TAU`
View	4% p.a., grades $(-1, -\frac{1}{2}, 0, \frac{1}{2}, 1)$	`VIEW`, `GRADES`
Active limit / TE budget	5% / 3% p.a.	`MAX_ACTIVE`, `MAX_TE`
Long-only	yes	`LONG_ONLY`
Trading costs (one-way)	50 / 25 / 10 bps	`COST_BPS`
Newey-West lags	6	–

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